Academic Statement of Purpose

of Sidharth (PhD Program in Computer Science and Engineering for Fall 2025)

Imagine sitting in a café, surrounded by conversations, yet effortlessly focusing on a single voice while tuning out the background noise. Despite today's advancements, hearabes still struggle to isolate a specific speaker in noisy environments. Understanding how the human brain performs this task could transform hearing technologies, such as earbuds, headphones, and hearing aids. I aim to develop next-generation hearing technologies by applying insights from how the brain focuses on a single speaker in a crowd, leveraging my background in audio machine learning and computational neuroscience to improve hearing aids for individuals with hearing impairments. This vision excites me, and I am eager to pursue it in the Computer Science and Engineering department at the University of Michigan. My focus will center on advancing speech and audio technology to create transformative listening experiences—developing hearable algorithms that leverage insights from the brain.

My experience in audio machine learning and computational neuroscience has shaped an interdisciplinary approach to extracting meaningful information from complex data. This has fueled my commitment to developing assistive technologies for individuals with hearing impairments and addressing real-world speech processing challenges.

DISPLACE Challenge 2023: As a student, I often found myself struggling to fully comprehend online lectures during pandemic due to the variety of accents and voices. I imagined how beneficial it would be to have a system that could identify and annotate each speaker in real-time, providing immediate, clear transcripts for easier comprehension. This need for accessible, real-time speech understanding motivated me to pursue my first project in speech processing at the Learning and Extraction of Acoustic Patterns (LEAP) lab at the Indian Institute of Science (IISc).

To build a baseline speaker diarization model, I conceptualized it can be solved through simple linear algebra techniques. Could different speaker signals be treated as independent components separable by matrix inversion? However, I quickly realized the limitations of basic matrix inversion due to data non-linearity and extensive code-switching. A discussion with my advisor introduced me to Independent Component Analysis (ICA), a powerful method for signal separation in both speech processing and neuroscience. This insight motivated me to explore ways of disentangling complex signals and made me understand the complexity of the problem.

Identifying pain biomarker from ECoG signals: While my work in speaker diarization introduced me to the challenges of signal separation, my growing interest in disentangling complex signals led me to explore similarly complex electrophysiological data, which poses unique challenges due to its high dimensionality. This interest brought me to collaborate with Dr. Jeffrey Herron, who was working on decoding cognitive behaviors from such data. Jeff's project involved a dataset of Electrocorticography (ECoG) signals recorded from epilepsy patients, post-surgery. Our goal was to identify neural biomarkers for pain, aiming to develop personalized treatments for chronic pain sufferers and improve communication for paralyzed patients by enabling them to express their pain more effectively. Having prior experience with electrophysiological signals from my undergraduate research, where I classified emotions from EEG data using coherence features, I was eager to dive in. However, finding biomarkers for pain presented a unique challenge—unlike emotions, pain isn't localized to a specific subcortical region but involves complex interactions across cortical areas. ICA, which is typically used to separate multivariate signals into additive, statistically independent components, could not handle this complexity due to the interdependence of signals across electrodes. I explored single-channel and multi-channel features for pain classification, with multi-channel features yielding the best results for both binary and ternary classifications. This approach enabled us to identify patient-specific electrode combinations and frequency bands for pain classification. This work was submitted to AAAI 2025 Workshop on Health Intelligence.

Through this project, I gained a deeper understanding of the complexities of decoding subjective experiences from neural signals—insights that closely aligned with the challenges I would later face in my work with speech technologies. The need to disentangle and enhance specific signals amidst complex backgrounds is a recurring theme across both pain detection and auditory signal enhancement.

Masked Audio Reconstruction and Targeted Speech Extraction: My growing interest in overcoming the challenges of isolating target speech from complex, dynamic noise environments led me to collaborate with Dr. Shyam Gollakota, who was working on developing innovative speech enhancement techniques for real-world applications.
Current hearables often struggle to adapt to varying acoustic environments, especially challenging ones like cathedral rooms, large reverberant spaces, or environments with sudden background noise changes. Our research project aimed to tackle this issue by developing a method to reconstruct audio signals at a specific microphone location using recordings from other nearby microphones, without requiring reference audio from the target location. We aimed to develop a physics-based, self-supervised model capable of learning the inter-transfer function between microphones, enabling it to adapt to varying acoustic environments. This approach has the potential to significantly improve the adaptability and personalization of hearing aids. To understand why these models perform poorly, I began addressing this issue by building a synthetic dataset to isolate and explore where the model's limitations lie. After extensive testing, I observed that the model struggled to reconstruct masked channel audio without a reference Room Impulse Response (RIR), as the predictions depended more on simulated convolution than independent inference. This insight led us to shift our focus toward a related challenge: isolating volume-controlled target speech from a noisy conversational mixture. In this approach, we used an embedding extractor network to identify the target speaker's embedding from the noisy mixture, while a target speech extractor module uses this embedding along with a volume control factor to output the desired speech volume. This technique promises applications in hearing aids, allowing users to adjust the target speaker's volume while minimizing background noise.

Speech Denoising: While experimenting with these models in speech technologies, I encountered the challenge of their high computational demands, especially for real-time applications such as hearing aids. This led me to reflect on how the human brain efficiently processes complex tasks with minimal computational resources. Intrigued by this, I began exploring ways to develop models that can operate effectively on edge devices, consuming low power while still delivering strong performance. Could we develop a physics-based model that understands the underlying audio rather than relying heavily on data-driven methods? Motivated by this question, I pursued a Machine Learning Researcher Internship at BrainChip in the summer of 2024. State-space models are known for their ability to capture long-range dependencies in sequences effectively, while also offering subquadratic training and inference time complexity. This made them an ideal choice for speech enhancement tasks, where not only capturing context over time is crucial but also computational efficiency is a priority for real-time applications. As I developed the state-space speech enhancement model, "aTENNuate", Initially, I trained the model with a signal error-based loss function, but this introduced artifacts like low-amplitude robotic noise and a muffled voice quality due to the loss of high-frequency details. After analyzing the spectrogram, I identified these issues and introduced a perceptual component to the loss function, which improved the model's ability to preserve speech quality. This successfully eliminated the artifacts, and aTTENuate outperforms current state-of-the-art real-time denoising models. This work has been submitted to IEEE ICASSP 2025.

Research Interests: I aim to develop next-generation hearables that enhance communication and deepen our understanding of neural processing by exploring models inspired by brain function, particularly in how the brain interprets sound. By integrating multimodal signals, like audio-visual cues, I envision real-time adaptive systems that adjust to users' needs, transforming assistive technologies to improve speech comprehension and overall listening experiences.
Pursuing PhD is a logical next step for me. During my PhD, I aim to tackle four key questions that are central to advancing hearable technologies. How can we develop adaptive, environment-aware algorithms for hearables that optimize speech enhancement in real time by adjusting to complex and unpredictable acoustic environments? Can we design low-computation, high-performance models that extract target speech in real time, even with limited frequency bandwidth and compressed audio inputs? How can multimodal inputs, such as audio and visual cues, be effectively integrated to improve target speech extraction in noisy environments? And, how can auditory attention decoding be integrated into real-time algorithms to continuously isolate and enhance target speech in fluctuating noise conditions? My interest in developing assistive audio technologies aligns with the work of **Dr. Dhruv Jain** and

Dr. Emily Mower Provost. I am particularly drawn to **Dr. Jain's Soundability Lab**, which focuses on advancing auditory technologies that enhance how people experience and interact with sound. This aligns perfectly with my vision of modeling human auditory behavior and creating next-generation hearables to enhance communication for all individuals. Additionally, I am excited about Dr. Provost 's research in mood classification and emotion recognition from speech, which complements my experience in decoding neural biomarkers for pain. I am eager to collaborate on projects that explore the development of real-time systems in next-generation hearables, enabling them to dynamically adapt to users' needs and environments, ultimately leading to more personalized and intuitive assistive technologies. Collaborating with them would allow me to investigate innovative solutions that cater to the diverse needs of users in real-world situations.

Why University of Michigan? The University of Michigan (UMich) attracts me to pursue a PhD, primarily for two reasons: the exceptional faculty whose research closely aligns with my interests, and UMich's strong collaborative ties with the medical department. I am excited about the prospect of interdisciplinary collaboration with Michigan Medicine, especially Dr. Devin McCaslin, whose research is focused on advancing hearing technologies and enhancing the understanding of auditory experiences. This potential collaboration would enable me to develop advanced hearable technologies that support not only people with hearing impairments but also enhance auditory experiences for all users in varied environments. Following my PhD, I aspire to stay in academia and work alongside like-minded colleagues to tackle interdisciplinary challenges. With my background in speech processing and computational neuroscience, I believe I can both benefit from and contribute to UMich's CSE research program, and I am excited about the prospect of furthering my education there.